WannierBSE: User Guide

Version1.0

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1. Introduction

1.1 Getting Help

The latest release of WannierBSE and the associated user manual can be found at the official website: https://quantum.web.nycu.edu.tw/wannierbse/

Alternatively, the manual and the development version of the source code may be cloned or downloaded from the official repository on GitHub: https://github.com/WannierBSE

For technical issues, bug reports, or questions regarding the functionality of WannierBSE, users are encouraged to utilize the official GitHub repository. You may raise specific issues or start discussions directly on the platform.

1.2 Citation

To support the continued development and maintenance of WannierBSE, we kindly ask that you cite the following references in any publication resulting from the use of this software.

Primary Reference

For the general use of the WannierBSE code, please cite:

[Ref] Guan-Hao Peng, Ping-Yuan Lo, Wei-Hua Li, Yan-Chen Huang, Yan-Hong Chen, Chi-Hsuan Lee, Chih-Kai Yang, and Shun-Jen Cheng, *"Distinctive Signatures of the Spin- and Momentum-Forbidden Dark Exciton States in the Photoluminescence of Strained WSe2 Monolayers under Thermalization",* **Nano Lett.** 19, 4, 2299–2312 (2019). https://doi.org/10.1021/acs.nanolett.8b04786

BibTeX entry:

```
@article{Peng2019,
  title = {Distinctive Signatures of the Spin- and Momentum-Forbidden
       Dark Exciton States in the Photoluminescence of Strained WSe2
       Monolayers under Thermalization},
  author = {Peng, Guan-Hao and Lo, Ping-Yuan and Li, Wei-Hua and Huang,
       Yan-Chen and Chen, Yan-Hong and Lee, Chi-Hsuan and Yang, Chih-
       Kai and Cheng, Shun-Jen},
  journal = {Nano Lett.},
  volume = {19},
  issue = {4},
  pages = {2299-2312},
  year = {2019},
  doi = {10.1021/acs.nanolett.8b04786}
}
```

Non-local Dielectric Function

If you utilize the semiclassical Poisson solver to calculate the non-local dielectric function, please also cite:

[Ref] Wei-Hua Li, Jhen-Dong Lin, Ping-Yuan Lo, Guan-Hao Peng, Ching-Yu Hei, Shao-Yu Chen, and Shun-Jen Cheng, "*The Key Role of Non-Local Screening in the Environment-Insensitive Exciton Fine Structures of Transition-Metal Dichalcogenide Monolayers*", **Nanomaterials** 13, 11, 1739 (2023). https://doi.org/10.3390/nano13111739

BibTeX entry:

```
@article{Lin2023,
 title = {Essential role of momentum-forbidden dark excitons in the
       energy transfer responses of monolayer transition-metal
       dichalcogenides},
 author = {Lin, Jhen-Dong and Lo, Ping-Yuan and Peng, Guan-Hao and Li,
       Wei-Hua and Huang, Shiang-Yu and Chen, Guang-Yin and Cheng,
       Shun-Jen},
 journal = {npj 2D Mater. Appl.},
 volume = {7},
 issue = {1},
 pages = {51},
 year = {2023},
 doi = {10.3390/nano13111739}
}
```

1.3 Credits

The WannierBSE software package was developed by the research group of Prof. Shun-Jen Cheng at National Yang Ming Chiao Tung University (NYCU). The project is supervised and guided by Prof. Shun-Jen Cheng.

Core Developers:

- Dr. Ping-Yuan Lo
- Dr. Wei-Hua Li
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1.4 License

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2. Theoretical Framework

WannierBSE is an efficient computational package for solving the Bethe-Salpeter equation (BSE) for neutral exciton in 2D materials within the Wannier tight-binding (WTB) framework based on the density-functional theory (DFT), implemented in the MATLAB language. The theoretical foundation of the BSE within this framework is based on the many-body formalisms developed by the seminal works of Refs. [1-7]. Technically, it takes advantage of global adaptive integration algorithm for fast and accurate construction of electron-hole (e-h) Coulomb kernels of BSE.

The core of the package relies on a multi-scaled methodology that integrates a microscopic Hamiltonian based on Wannier functions [8,9] with a macroscopic Poisson solver to calculate the screened Coulomb interaction.

WannierBSE can efficiently construct the BSE for neutral exciton in 2D materials based on the DFT-calculated band structures and Bloch wavefunctions, and dielectric screening from the DFT+GW or macroscopic electromagnetic theory. This flexibility enables WannierBSE to perform multiscale simulations of exciton spectra in 2D materials, capturing the effects of realistically complex dielectric environments.

The WannierBSE package is developed to numerically solve the Bethe-Salpeter equation (BSE) [3,10,11],

$$\begin{aligned}(\epsilon_{c\boldsymbol{k}+\boldsymbol{q}} - \epsilon_{v\boldsymbol{k}})A_{S\boldsymbol{Q}}(vc\boldsymbol{k}) \\ - \sum_{v'c'\boldsymbol{k}'} \big[W_{\boldsymbol{Q}}^{eh,d}(vc\boldsymbol{k}, v'c'\boldsymbol{k}') \\ - V_{\boldsymbol{Q}}^{eh,x}(vc\boldsymbol{k}, v'c'\boldsymbol{k}')\big]A_{S\boldsymbol{Q}}(v'c'\boldsymbol{k}') = E_{S\boldsymbol{Q}}A_{S\boldsymbol{Q}}(vc\boldsymbol{k}),\end{aligned} \tag{2.1}$$

where $\epsilon_{\mathrm{n}k}$ is the kinetic energy, E_{SQ} is the energy of exciton state labeled by quantum number S with center-of-mass (CoM) wave vector $\boldsymbol{Q}$. The e-h direct and exchange Coulomb kernels read [5,6,12,13],

$$\begin{aligned}W_{\boldsymbol{Q}}^{eh,d}(vc\boldsymbol{k}, v'c'\boldsymbol{k}') \\ = \int\int d^3\boldsymbol{r}_1 d^3\boldsymbol{r}_2\, \psi_{c\boldsymbol{k}+\boldsymbol{Q}}^*(\boldsymbol{r}_1)\psi_{v'\boldsymbol{k}'}^*(\boldsymbol{r}_2)W(\boldsymbol{r}_1,\boldsymbol{r}_2)\psi_{v\boldsymbol{k}}(\boldsymbol{r}_2)\psi_{c'\boldsymbol{k}'+\boldsymbol{Q}}(\boldsymbol{r}_1),\end{aligned} \tag{2.2}$$

$$\begin{aligned}V_{\boldsymbol{Q}}^{eh,x}(vc\boldsymbol{k}, v'c'\boldsymbol{k}') \\ = \int\int d^3\boldsymbol{r}_1 d^3\boldsymbol{r}_2\, \psi_{c\boldsymbol{k}+\boldsymbol{Q}}^*(\boldsymbol{r}_1)\psi_{v'\boldsymbol{k}'}^*(\boldsymbol{r}_2)V(\boldsymbol{r}_1-\boldsymbol{r}_2)\, \psi_{c'\boldsymbol{k}'+\boldsymbol{Q}}(\boldsymbol{r}_2)\psi_{v\boldsymbol{k}}(\boldsymbol{r}_1).\end{aligned} \tag{2.3}$$

Here, $\psi_{\mathrm{n}\boldsymbol{k}}(r)$ is the single-electron wavefunction, $W(\boldsymbol{r}_1,\boldsymbol{r}_2) = \int \mathrm{d}^3\mathbf{r}'\varepsilon^{-1}(\mathbf{r}_1,\mathbf{r}')V(\boldsymbol{r}'-\boldsymbol{r}_2)$ is the screened Coulomb potential, with $\varepsilon^{-1}(\boldsymbol{r}_1,\boldsymbol{r}')$ being the inverse non-local dielectric function and $V(\mathbf{r}) = \frac{e^2}{4\pi\varepsilon_0|\mathbf{r}|}$ being the bare Coulomb potential.

- **Note on Version 1.0:** In the current release of WannierBSE (v1.0), the electron-hole exchange interaction (Eq. 2.3) is neglected. The current implementation

focuses on the direct interaction term, which dominates the exciton binding energy. Future versions of the package are planned to incorporate the exchange kernel to address fine-structure splitting.

By formulating the e-h Coulomb kernels in the WTB framework, which unitarily connects the single-electron Bloch wavefunctions with the maximally localized Wannier function (MLWF) basis $W_{j\boldsymbol{R}}(\boldsymbol{r})$ by [14]

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N}} \sum_{j} \sum_{\boldsymbol{R}} C_j^{(n)}(\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{R}} W_{j\boldsymbol{R}}(\boldsymbol{r}), \tag{2.4}$$

where j labels the orbital-like WFs, $\boldsymbol{R}$ indicates the unit cell that the WF locates at, and the coefficients $C_j^{(n)}(\boldsymbol{k})$ are determined by the WTB Hamiltonian

$$H_{ij}^{WTB}(\boldsymbol{k}) = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}} \langle W_{i\boldsymbol{0}} | \widehat{H}^{KS} | W_{j\boldsymbol{R}} \rangle, \tag{2.5}$$

which is equivalent to the Kohn-Sham Hamiltonian $\widehat{H}^{KS}$ represented in the WF basis. Within the WTB framework, the Bloch wavefunctions $\psi_{n\boldsymbol{k}}(\boldsymbol{r})$ can be expanded by a small set (typically 10^1~10^2) of MLWFs, significantly reduces the computational costs of constructing e-h Coulomb kernels of the BSE. For further technical details, see Peng et al., *Nano Lett.* 2019, 19, 2299–2312 [14].

2.1 Dielectric Screening

The accurate description of excitonic properties in TMD-MLs is fundamentally linked to the treatment of dielectric screening. The methodology employed here for modeling non-local screening and the dielectric environment of 2D semiconductors builds upon the influential work of Ref. [5,6,13,15,16]. Due to reduced dimensionality, TMDs exhibit weak intrinsic screening, leading to enhanced e-h interactions. In WannierBSE, the dielectric environment is treated by solving the Poisson equation for a layered structure, accommodating encapsulated or substrate-supported monolayers. This approach allows for the accurate description of excitonic complexes embedded in multi-layered dielectric environments, accounting for the non-local screening effects arising from the surrounding media.

The screened Coulomb interaction $W(\boldsymbol{r}_1, \boldsymbol{r}_2)$ is essentially the Green's function describing the potential at $\boldsymbol{r}_1$ due to a point charge at $\boldsymbol{r}_2$. In the presence of a non-local dielectric function $\varepsilon^{-1}(\boldsymbol{r}_1, \boldsymbol{r}')$, the interaction obeys the general form of the Poisson equation [6,13,16-18]:

$$\boldsymbol{\nabla}_{\boldsymbol{r}_1} \cdot \int \mathrm{d}^3\mathbf{r}' \varepsilon^{-1}(\mathbf{r}_1, \mathbf{r}') \boldsymbol{\nabla}_{\boldsymbol{r}'} W(\boldsymbol{r}', \boldsymbol{r}_2) = -\frac{\mathrm{e}^2}{\varepsilon_0} \delta(\boldsymbol{r}_1 - \boldsymbol{r}_2). \tag{2.6}$$

To exploit the lattice translational symmetry of the 2D plane, the equation is reformulated using an in-plane Fourier expansion. Under a long-range approximation, the system is modeled as a vertical heterostructure where the dielectric environment and the TMD layer are treated as distinct regions with specific boundary conditions.

WannierBSE employs a piece-wise dielectric function to model the vertical heterostructure. A typical configuration consists of a TMD monolayer of effective thickness d embedded in a dielectric environment (e.g., air, hBN, or SiO_2). Within the monolayer region ($-d/2 < z < d/2$), the screening is characterized by a q-dependent bulk dielectric function, which in in the random-phase approximation is given by [15,16]:

$$\varepsilon_{\mathrm{TMD}}(q) = 1 + \frac{1}{(\varepsilon_{2D}-1)^{-1} + \alpha\frac{q^2}{q_{TF}^2} + \left(\frac{\hbar^2 q^2}{2m_0 E_{pl}}\right)^2}. \tag{2.7}$$

In this expression, ε_{2D} is the in-plane dielectric constant of the bulk TMD, α is a dimensionless fitting parameter, q is the in-plane momentum, and E_{pl} represents the plasma peak energy. The non-locality of the screening is further governed by the Thomas-Fermi screening wave vector, q_{TF}, which is mathematically defined as [13,16]:

$$q_{TF} = \sqrt{\frac{e^2 m_0}{\pi^2 \epsilon_0 \hbar^2}\left(\frac{3\pi^2 \varepsilon_0 m_0 E_{pl}^2}{e^2 \hbar^2}\right)^{\frac{1}{3}}}. \tag{2.8}$$

The resulting effective dielectric function of the system is defined as [5,6]

$$\varepsilon(\boldsymbol{q}) = \frac{\overline{V}(\boldsymbol{q})}{\overline{W}(\boldsymbol{q})}. \tag{2.9}$$

Here, $\overline{V}(\boldsymbol{q})$ represents the averaged unscreened interaction, and $\overline{W}(\boldsymbol{q})$ is the effective 2D screened potential averaged over the thickness d. The coefficients required to determine the final potential are calculated by matching boundary conditions at the interfaces, ensuring the continuity of the potential and the displacement field across the heterostructure boundaries. For a complete derivation of these expressions and the specific parameters for various TMD materials, see Li et al., *Nanomaterials* **2023**, *13*, 1739 [19].

3. WannierBSE: Software Implementation

WannierBSE is designed for seamless integration with Wannier90 calculations and features a built-in macroscopic Poisson solver to determine the non-local dielectric screening function. While highly automated, WannierBSE is engineered for flexibility, allowing users to import their own custom models into the computational pipeline. To accommodate different research needs, the package supports two distinct operational workflows:

1. Fully Automated Mode: Users provide only the fundamental Wannier Hamiltonian (wannier90_hr.dat), and the package automatically resolves the required band structure, momentum grid, and dielectric screening parameters.

2. Custom/User-Defined Mode: Users may bypass internal solvers by providing their own pre-calculated Tight-Binding models, customized k-meshes, or specialized dielectric functions.

To manage these workflows effectively, WannierBSE utilizes a modular directory structure designed to strictly separate user configurations, raw input data, and cached results. The directory architecture is organized as follows:

- Parameters: Contains the six mandatory control files that define the physical and numerical environment: control.txt, structure.txt, WF_centers.txt, kmesh_control.txt, WTB_control.txt, and dielectric_control.txt.
- User_input: The primary gateway for external data. Users should place wannier90_hr.dat here for automated generation. Alternatively, users can provide their own pre-calculated files (Bands, K-mesh, or Dielectric) to override internal solvers.
- Precomputed_data: Acts as the software's internal cache. This is where the code stores generated files, such as kmesh_WBSE.txt, v*_TB_WBSE.txt, and epsilon_WBSE.txt, to accelerate subsequent simulations.
- Exciton_data: The final repository for all physical results. This folder centralizes the calculated excitonic eigenvalues (Ex.mat), eigenvectors (A.mat), momentum grids (k.mat), and execution benchmarks/log files.
- Functions_for_BSE: Contains the core MATLAB kernels, libraries, and numerical routines required for the calculation.

The software manages the workflows through an intelligent Data Resolution Priority. Upon execution, the solver first scans the User_input directory for existing physical data (Bands, K-mesh, or Dielectric files). If these files are detected, the internal solvers are bypassed, and the user-provided data is ingested directly. If the User_input directory is empty, the software automatically looks for the corresponding data in the Precomputed_data folder saved as cache. Finally, if no cached data is found, the software triggers the generation engines and stores the resulting data in the Precomputed_data cache for future efficiency.

3.1 Parameters

The execution of WannierBSE is governed by a set of plain text input files that must be present in the working directory. These files allow the user to define the physical system, the computational grid, the dielectric screening properties, and the specific algorithmic controls for the BSE calculation.

The input system is designed to be modular. While some files are strictly required for every calculation, others are optional or conditional—used only to generate intermediate data (such as the k-mesh or dielectric function) when pre-calculated files are unavailable.

Mandatory Input Files Every standard WannierBSE calculation requires the following files to be present:

- control.txt: The primary configuration file for general simulation settings, active band windows, and parallelization.
- structure.txt: Defines the crystal structure and lattice parameters in the standard Wannier90 format.
- WF_centers.txt: Specifies the spatial mapping between Wannier functions and atomic centers.

Conditional & Optional Files The following files are required only under specific conditions:

- WTB_control.txt: Required when using wannier90_hr.dat to construct the Hamiltonian. It identifies the top valence band and the lowest conduction band indices.
- kmesh_control.txt: Required only if a kmesh.txt file is not already present. It defines parameters for generating a non-uniform k-point grid.
- dielectric_control.txt: Required only if an epsilon.txt file is not already present. It defines the layer architecture for the internal dielectric function generator.

In the following subsections, the syntax and required parameters for each file are described in detail.

3.1.1 General Control: control.txt

This file defines the fundamental computational settings for the BSE calculation, including the active band window and parallelization preferences. Please note that the MATLAB Parallel Computing Toolbox is required to run these calculations.

Parameter Description:

Keyword	Type	Description
d	Real	The effective thickness of the 2D material slab, specified in Angstroms (Å). This value is used for the truncation of the Coulomb interaction in the direction perpendicular to the layer.
NVAL	Integer	The number of lowest-energy exciton eigenstates (eigenvalues) to compute during the diagonalization of the BSE Hamiltonian.
Nc	Integer	The number of conduction bands to include in the electron-hole basis set. *(Note: This must be less than or equal to the number of c*_TB.txt input files provided).*
Nv	Integer	The number of valence bands to include in the electron-hole basis set. *(Note: This must be less than or equal to the number of v*_TB.txt input files provided).*
Number_par	Integer	The number of parallel workers (threads) to utilize for the calculation. This controls the size of the MATLAB parallel pool. *(Note: This value should not exceed the number of physical CPU cores available on the local machine. For low-memory systems or laptops, setting this value to 1 will disable the parallel pool to conserve resources.)*

Example File:

```
d = 6.26              % 2D material thickness in Angstroms
%-----------------------BSE calculation options----------------------------
NVAL = 100          % Number of exciton eigenvalues to find
Nc = 2                % Number of conduction bands in BSE
Nv = 2                % Number of  valence bands in BSE
%-------------------parallel calculation parameters------------------------
Number_par = 6    % Number of parallel threads
```

3.1.2 Structure Definition: structure.txt

This file defines the crystal lattice vectors and the atomic positions of the system. The format strictly follows the conventions used by the Wannier90 software package, facilitating easy migration of existing input data.

Cell Lattice Vectors The lattice vectors are specified in Cartesian coordinates enclosed within the *unit_cell_cart* block. The components must be provided in Angstroms (Å). The block format is as follows:

begin unit_cell_cart

A_x^1 A_y^1 A_z^1

A_x^2 A_y^2 A_z^2

A_x^3 A_y^3 A_z^3

end unit_cell_cart

In this configuration, the values A_x^1, A_y^1, A_z^1 correspond to the Cartesian components of the first lattice vector $\mathbf{A_1}$ with subsequent rows corresponding to vectors $\mathbf{A_2}$ and $\mathbf{A_3}$, respectively.

Ionic Positions Similarly, the atomic positions are defined using the atoms_cart block. These coordinates must also be specified in Angstroms (Å) and absolute Cartesian coordinates.

begin atoms_cart

Label R_x R_y R_z

...

end atoms_cart

The first entry on each line, *Label,* represents the atomic symbol (e.g., Mo, S). The following three entries $\boldsymbol{R} = (R_x, R_y, R_z)$ denote the position coordinates of that atom in the Cartesian frame.

Example File:

```
begin unit_cell_cart
         1.5915472948    2.7566392707    0.0000000000
        -1.5915472948    2.7566392707    0.0000000000
         0.0000000000    0.0000000000   15.1264784281
end unit_cell_cart

begin atoms_cart
   Mo     0.0000000000    3.6755187193    7.5632392140
   S      0.0000000000    1.8377596680    9.1267531759
   S      0.0000000000    1.8377596680    5.9997252522
end atoms_cart
```

3.1.3 Wannier Function Centers: WF_centers.txt

This file establishes the spatial mapping between the Wannier functions and their respective atomic centers. It is essential for associating each orbital-like Wannier function with a specific ion in the unit cell.

The file contains two columns of integers. The first column represents the index i of the Wannier function, WF_Index(i). The second column specifies the index I of the atomic center τ_I where the i-th Wannier function is localized.

The atomic index I corresponds strictly to the order in which atoms are listed in the *atoms_cart* block of the input structure (found in the original wannier90.win file). For

example, an index of *I=1* refers to the first atom listed immediately after the begin *atoms_cart* line, *I=2* refers to the second atom, and so forth.

Example File:

```
%  WF_Index(i) Atom_Index(I)

      1             2
      2             2
      …             …
      7             3
      8             3
      …             …
     21             1
     22             1
```

3.1.4 K-Mesh Generation: kmesh_control.txt

This configuration file defines the parameters for generating the momentum space grid (k-mesh) used in the BSE calculation. The WannierBSE code invokes the mesh generation routine only if a pre-existing kmesh.txt file is not found in the working directory. If kmesh.txt is already present, this control file is ignored.

The generation algorithm constructs a non-uniform grid characterized by a dense region and a coarse region, see Figure 1. The base resolution of the global Brillouin zone is defined by Nk_coarse, which sets the geometric split along the line from the Γ point to the K point. This parameter effectively determines the mesh fineness for the bulk of the Brillouin.

The parameter Dense_range acts as an integer factor to determine the coverage of the dense region in units of the coarse grid. It specifies the spatial extent of high-resolution sampling centered at the high-symmetry K and K′ points. To maintain geometric consistency and avoid overlapping points between adjacent valleys, this value must satisfy the constraint, $\text{Dense_range} \leq \text{Nk_coarse}/2$. The internal grid density within these refined regions is governed by Nk_dense, which represents the number of k-points sampled along a single direction within the dense sub-mesh.

Example File:

```
% ---  Control parameters for k-mesh generation  ---

Nk_coarse= 6        % Number of k-points along one direction in the
                      coarse region.

Nk_dense = 4        % Number of k-points along one direction in the
                      dense region.

Dense_range = 2     % Integer factor to determine dense region size
                      in units of the coarse grid.
```

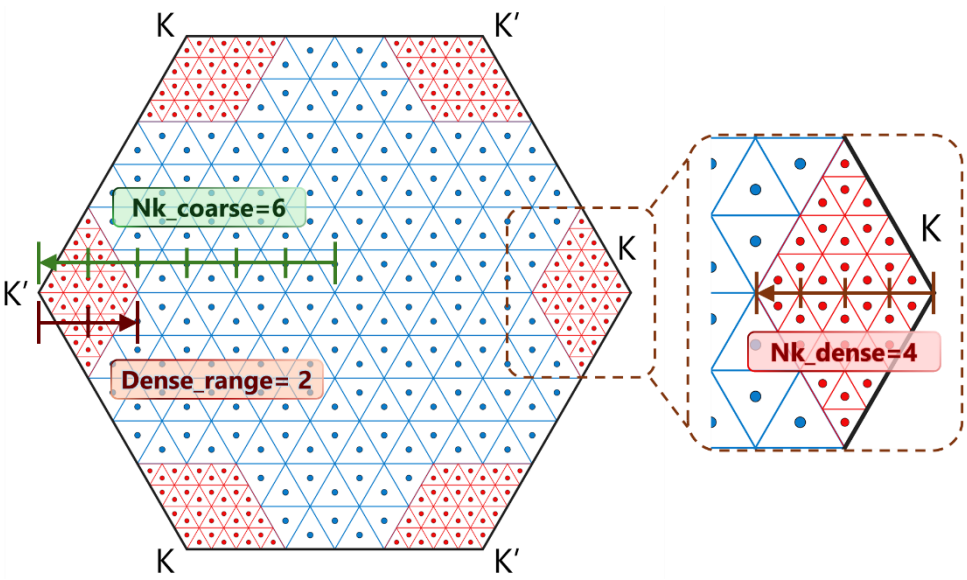


Figure 1. Schematic of the non-uniform k-mesh. The Brillouin zone is partitioned into a coarse grid (blue) for the global domain and a high-density grid (red) localized around the K and K′ valleys. The overall coarse resolution is determined by Nk_coarse = 6, which sets the geometric split along the line from the Γ point to the K point. The parameter Dense_range = 2 acts as an integer factor to determine the coverage of the dense region in units of the coarse grid, as illustrated, it allocates two-sixths (2/6) of the total the Γ to the K distance to the high-resolution zones. The internal sampling density within these refined regions is controlled by Nk_dense = 4, allowing for enhanced precision at high-symmetry points while maintaining computational efficiency throughout the remaining Brillouin zone.

3.1.5 Wannier Tight-Binding Control (WTB_control.txt)

This file is required when the system Hamiltonian is constructed using the wannier90_hr.dat hopping parameter file generated by Wannier90. Since the Wannier90 output often contains a large number of bands, it is necessary to explicitly identify the indices that define the semiconductor band gap.

Optical transitions are most likely to occur between the bands in the immediate vicinity of the band gap. Therefore, the parameters in this file serve as reference points for the band selection process. The parameter VBM (Valence Band Maximum) identifies the index of the top-most valence band, while CBM (Conduction Band Minimum) identifies the index of the lowest conduction band.

Example File:

```
% Control parameters for Wannier Tight-Binding Calculation

VBM = 14            % VBM: Valence Band Maximum index
CBM = 15            % CBM: Conduction Band Minimum index
```

3.1.6 Dielectric Model Control: dielectric_control.txt

In the absence of a pre-calculated epsilon.txt file, WannierBSE utilizes an internal Poisson solver to generate the necessary dielectric function. This file provides the geometric and material parameters required to construct the dielectric environment model.

The model assumes a multi-layer heterostructure. The code constructs the stack based on the number of layers defined above (t) and below (s) the central TMD monolayer, see Figure 2.

- Total Layers: The total number of dielectric regions, including the TMD monolayers, is given by $N_{total} = t + 1 + s$.
- Boundaries: There must be $s + t$ interfaces (boundaries) defined to separate these regions.
- Ordering: All lists (dielectric constants and boundaries) must be specified from Top to Bottom (positive z to negative z).

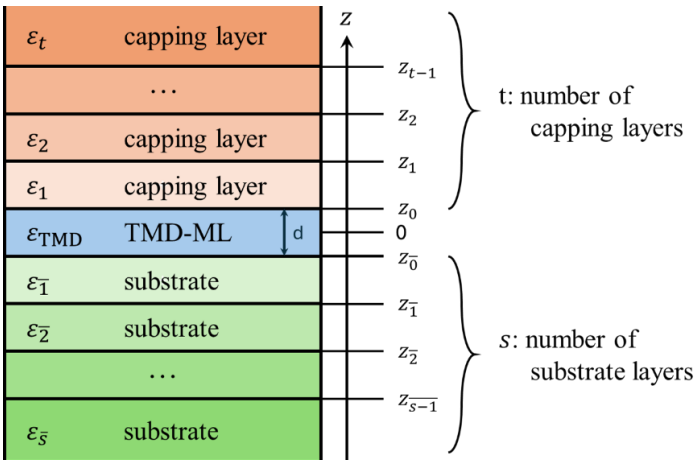


Figure 2: Schematics of the multilayer structure with TMD embedded in the middle. The TMD-ML of thickness d is embedded within a stack of capping t and substrate s layers, resulting in a spatially dependent, piecewise dielectric function. The electronic environment is characterized by the q-dependent bulk dielectric function of the monolayer, $\varepsilon_{\mathrm{TMD}}(q)$, while the surrounding media are defined by their respective dielectric constants, ε_t for the capping layers and $\varepsilon_{\bar{s}}$ for the substrate layers. The vertical z-coordinates (z_n) denote the discrete interfaces between adjacent dielectric regions.

Parameter Description:

Keyword	Type	Description
alpha	Real	A dimensionless fitting parameter associated with the Rytova-Keldysh potential.
Epl	Real	The plasmon peak energy, specified in eV.
ep_p	Real	The specific in-plane dielectric constant of the TMD monolayer.
t_layers	Integer	The number of dielectric layers situated above the TMD monolayer.
s_layers	Integer	The number of dielectric layers situated below the TMD monolayer.
t_dielectric	List	A comma-separated list of dielectric constants for the layers above the TMD monolayer. Following the convention in Figure 2, the list is ordered starting from the top layer: t_dielectric= ε_t , … , $\varepsilon_2, \varepsilon_1$. ** This list must contain exactly t_layers values.*
s_dielectric	List	A comma-separated list of dielectric constants for the layers below the TMD monolayer. Following the convention in Figure 2, the list is ordered starting from the layer closest to the TMD: s_dielectric= $\varepsilon_{\bar{1}}$, $\varepsilon_{\bar{2}}$, … , $\varepsilon_{\bar{s}}$. ** This list must contain exactly s_layers values.*
layer_boundaries	List	A comma-separated list of vertical coordinates (in Å), ordered from top to bottom, defining the planar interfaces between adjacent layers. **This list must contain exactly s + t values.*

Example File:

```
% Dielectric Function Control File

% --- Material Parameters for the TMD (SI units) ---

alpha = 1.55        % Dimensionless fitting parameter
Epl   = 22.5        % Plasmon peak energy (eV)
ep_p  = 13.24       % In-plane dielectric constant (static)

% --- Layer Architecture
% t: number of dielectric layers ABOVE the TMD
% s: number of dielectric layers BELOW the TMD
t_layers = 2
s_layers = 1

% --- Dielectric Constants of Surrounding Layers
% IMPORTANT:
% - Do NOT include the TMD monolayer here.
% Dielectric constants must be listed from top to bottom
```

```
t_dielectric = 1, 5.89
s_dielectric = 3.9

% --- Layer Boundaries (Top to Bottom)
% Units: Angstrom (Å)

layer_boundaries = 3.13, -3.13
```

3.2 Physical Data: User_input & Precomputed_data

The input handling in WannierBSE is designed for maximum flexibility, employing an intelligent "check-and-generate" workflow. This system allows the software to adapt to various research scenarios, whether you are relying on the built-in automated solvers or providing custom-tailored physical models. These files, including the system Hamiltonian, momentum-space grids, wavefunction coefficients, and dielectric screening, serve as the heavy numerical foundation upon which the BSE calculation is built.

To manage this workflow effectively, WannierBSE utilizes a dual-directory system to strictly separate user overrides from automated results:

- The User_input Folder: This is the primary directory for user-defined models. By placing files here, you "force" WannierBSE to use your specific data (e.g., a k-mesh from a different code or an experimental dielectric constant) instead of calculating them internally.

- The Precomputed_data Folder: This acts as the software's internal cache. When the internal engines run, they save their results here with a _WBSE suffix. This ensures that in subsequent runs, the code can quickly reload data from this cache rather than recalculating it.

Priority Logic & Dependencies:

When the simulation begins, the code scans these directories following a specific priority: User_input > Precomputed_data > Generation. If a required file is missing from both folders, WannierBSE automatically triggers the internal engines based on the following dependencies:

- Hamiltonian & Wavefunctions: The Wannier Hamiltonian (wannier90_hr.dat) must always be placed in the User_input folder. It serves as the "seed" for the generation of conduction (c*_TB_WBSE.txt) and valence (v*_TB_WBSE.txt) band files. In the case that the number of band files found does not match the N_c and N_v values specified in control.txt, the code will re-diagonalize the Hamiltonian on the fly to generate them.

- K-point Grid: The code first searches for kmesh.txt. If missing, the package automatically generates the grid based on the parameters specified in kmesh_control.txt.
- Dielectric Screening: If epsilon.txt is not provided, the package computes the screening function using the parameters defined in dielectric_control.txt.

Naming Conventions:

To ensure the software correctly identifies your files, you must follow these naming rules:

Data Type	User Input Name (User_input)	Automated Cache Name (Precomputed_data)
Hamiltonian	wannier90_hr.dat	*(N/A - Always in User_input)*
K-Mesh	kmesh.txt	kmesh_WBSE.txt
Bands	v*_TB.txt, c*_TB.txt	v*_TB_WBSE.txt, c*_TB_WBSE.txt
Dielectric	epsilon.txt	epsilon_WBSE.txt

Note on File Formats: The internal structure (columns and units) of these files is identical regardless of their location. For example, a v1_TB.txt provided by the user must follow the same column specifications as a generated v1_TB_WBSE.txt file.

3.2.1 Momentum Space Grid: kmesh.txt

This file defines the discretized momentum space (k-points) used for the calculation. To optimize computational efficiency while capturing the physics of excitons (which are typically localized in momentum space), WannierBSE employs a non-uniform mesh with D_{3h} symmetry.

Users have the flexibility to either manually provide this file or rely on the software's automatic generation capabilities. If kmesh.txt is not present in the working directory, WannierBSE will automatically construct the grid using the parameters specified in the kmesh_control.txt file (see Section 3.1.4). However, if the user manually supplies a kmesh.txt file, the internal generation step is bypassed, and the control parameters are ignored.

To ensure numerical stability, manually provided grids are automatically validated and reorganized so that all points belonging to the dense region are listed first, followed by those in the coarse region. This automated sorting ensures that user-defined meshes strictly adhere to the solver's internal requirements without needing manual reformatting.

The file consists of a five-line header containing mesh statistics, followed by the list of k-point coordinates.

- Header Block (Lines 1–5):
    1. Dense Region Range: The size of the dense region in units of the coarse grid.
    2. Coarse Divisions: The number of divisions along the vector path in the coarse region.
    3. Dense Divisions: The number of divisions along the vector path in the dense region.
    4. Total Coarse Points: The total count of grid points belonging to the coarse region.
    5. Total Dense Points: The total count of grid points belonging to the dense region.
- Coordinate Block (Line 6 onward):
  The remaining lines list the Cartesian coordinates of every k-point in the mesh. There are three columns corresponding to k_x, k_y, and k_z.

Example file:

```
  10              % Dense Region Range
  20              % Coarse Divisions
  10              % Dense Divisions
1200              % Total Coarse Points
1200              % Total Dense Points
-0.3289873897   0.6078110657   0.0000000000
-0.2960886507   0.6647933531   0.0000000000
...
```

3.2.2 Conduction and Valence Bands: c*_TB.txt / v*_TB.txt

These files contain the energy eigenvalues and the wavefunction coefficients (eigenvectors) in the Tight-Binding scheme for the specific bands involved in the BSE calculation.

The filenames use a lowercase prefix followed by the integer band index. This index corresponds to the specific conduction or valence band sorted by energy distance from the gap (e.g., 1 is the band closest to the gap).

- c*_TB.txt: Conduction band data (e.g., `c1_TB.txt`).
- v*_TB.txt: Valence band data (e.g., `v1_TB.txt`).

Similar to the grid file, users may provide these files manually or allow WannierBSE to generate them via internal helper functions.

It is important to notice that, if provided manually, the data in these files must correspond exactly to the k-points defined in `kmesh.txt`. The number of blocks and the

coordinate values must match line-by-line. However, as the k-mesh is automatically reorganized to prioritize the dense region, WannierBSE will dynamically rearrange the data blocks within the band files to ensure they are perfectly synchronized with the new momentum grid. This automated alignment maintains the physical integrity of the eigenvalues and eigenvectors without requiring the user to manually reorder the input files. For research traceability, these rearranged files are stored in the TB_Bands folder within the Exciton_data directory.

The file is structured as a sequence of data blocks. Each block corresponds to a single k-point in the mesh. If the basis set consists of N_{orb} Wannier functions, each block contains $N_{orb} + 2$ lines.

The block structure, repeated for every k-point, is as follows:

1. Line 1 ($\boldsymbol{k}$ -vector): The Cartesian coordinates k_x, k_y, k_z (in Å^{-1}).
2. Line 2 (Eigenvalue): The energy eigenvalue of the band $\epsilon_{n\boldsymbol{k}}$ (in eV).
3. Lines 3 to End (Eigenvectors): The coefficients of the eigenvector expanded in the Wannier Tight-Binding basis. There are N_{orb} lines. Each line contains three columns:

- $\mathrm{Re}[C_j(\boldsymbol{k})]$: Real part of the coefficient.
- $\mathrm{Im}[C_j(\boldsymbol{k})]$: Imaginary part of the coefficient.
- $\mathrm{Abs}[C_j(\boldsymbol{k})]$: Absolute magnitude.

Notation Alert: Users should be careful to distinguish between the band index and the eigenvector coefficient index, as both typically use the letter 'c'. The lowercase notation c_n refers specifically to the n-th conduction band, which is reflected in the filename (e.g., c1_TB.txt). In contrast, the uppercase notation C_j refers to the j-th component of the eigenvector, represented by the individual rows of data within that file.

Example file:

```
% --- Block for k_point 1 ---
 -0.328987   0.607811   0.000000    % k-vector (A^-1)
 -4.035566                          % Eigenvalue (eV)
 -0.161540  -0.166478   0.231970    % C_1 (Re, Im, Abs)
  0.004001   0.009269   0.010096    % C_2 (Re, Im, Abs)
 -0.223895  -0.030976   0.226027    % C_3 (Re, Im, Abs)
  ...
```

3.2.3 Dielectric Screening: epsilon.txt

This file contains the tabulation of the momentum-dependent static dielectric function, $\epsilon(|\boldsymbol{q}|)$, which describes the screening of the Coulomb interaction in the 2D material system.

Users may choose to provide a pre-calculated dielectric function obtained from other methods or use the internal model provided by WannierBSE. If the epsilon.txt file is not found in the working directory, the package automatically invokes its internal

Poisson solver to generate the screening data based on the layer architecture and material constants specified in dielectric_control.txt (see Section 3.1.6). Conversely, if the file is manually provided, the internal generation step is skipped, and dielectric_control.txt is ignored.

The file is formatted as a standard two-column ASCII text file. The first column represents the magnitude of the momentum transfer vector, $|\boldsymbol{q}|$, in units of Å^{-1}, covering the range $0 \leq |\boldsymbol{q}| \leq \infty$. The second column contains the corresponding dimensionless dielectric function value, $\epsilon(|\boldsymbol{q}|)$.

Example file:

```
%---      q(A-1)          epsilon(q)
       0.000000e+00     1.000000e+00
       2.500250e-04     1.009815e+00
       5.000500e-04     1.019619e+00
       7.500750e-04     1.029413e+00
       1.000100e-03     1.039197e+00
       1.250125e-03     1.048970e+00
       ...
```

3.2.4 Hamiltonian Matrix: wannier90_hr.dat

This file contains the Hamiltonian matrix elements calculated in the maximally localized Wannier function basis. It is a standard output generated by the Wannier90 software package.

Within the WannierBSE workflow, this file serves as the fundamental source data. If the processed wavefunction files (c*_TB.txt and v*_TB.txt) are not present in the input directory, the code reads wannier90_hr.dat to construct the Hamiltonian, diagonalize it at the required k-points, and generate the missing wavefunction files automatically. Users should simply copy this file from their Wannier90 output folder into the Precomputed_Physical_Data directory without modification.

The file follows the standard format defined by the Wannier90 developers, consisting of a header defining the system size followed by the list of matrix elements. For a detailed line-by-line description of the headers, degeneracy blocks, and column definitions, please refer to the official Wannier90 User Guide.

Example File:

```
written on 14 Jul 2025 at 10:15:35
      22
      91
3    2    1    1    2    2    1    1    1    1    1    2    3    1    1
1    1    1    1    1    1    3    1    1    1    1    1    1    1    1
...
   -6    3    0    1    1    0.000236   -0.000000
   -6    3    0    2    1    0.000000    0.000000
   -6    3    0    3    1    0.000000    0.000000
   -6    3    0    4    1    0.000005    0.000000
```

```
-6    3    0    5    1    0.000005   -0.000000
-6    3    0    6    1   -0.000000    0.000005
-6    3    0    7    1   -0.000270    0.000000
...
```

3.3 Exciton Data

This section describes the output files generated by the WannierBSE package. After the input configuration is finalized, the simulation is initiated by executing the WBSE.m main script. Upon successful completion, the package produces a suite of files containing the primary physical results and auxiliary simulation data.

The primary physical results are contained in two main files: Ex.mat, which stores the exciton energy spectrum, and A.mat, which contains the excitonic wavefunctions. In addition to these core results, the package generates three auxiliary log files, WBSE_log.txt, k.mat, and wi.mat, that record the simulation performance and the specific grid geometry used.

To ensure efficient storage and fast post-processing, the physical data is saved in MATLAB binary format (.mat). Detailed descriptions of the format and content of each file are provided in the following subsections.

3.3.1 Exciton Energy: Ex.mat

This file represents one of the main outputs of the WannierBSE package. It stores the fundamental results of the BSE diagonalization: the exciton energy eigenvalues, denoted as E_S^X.

The data is stored as a double-precision column vector. The eigenvalues are arranged in ascending order (from lowest energy to highest). The total number of entries corresponds to the parameter NVAL specified in the control.txt file. All energy values are provided in electron-volts (eV).

Example file:

```
                    E_S^X
                1.1180  % E_1^X
                1.1187  % E_2^X
                1.1455  % E_3^X
                1.1455  % E_4^X
                1.2652  % E_5^X
...
```

3.3.2 Exciton Wavefunctions: A.mat

Together with Ex.mat, this file constitutes the core output of the WannierBSE package. It contains the excitonic envelope functions (eigenvectors), denoted as $A_S(vck)$, which describe the probability amplitude of the electron-hole pair in reciprocal space.

The data is stored as a 2D matrix of complex double-precision numbers.

- Columns (Exciton States): Each column corresponds to a specific exciton eigenstate S, sorted by energy in ascending order.
- Rows (Basis Index): This index is defined by the specific combination of the valence band index (v_i), the conduction band index (c_j), and the momentum vector (k_l).

The elements of the eigenvector are organized hierarchically. The file lists the coefficients for all k-points for the first band pair (v_1, c_1), followed by the next conduction band (v_1, c_2), continuing until all conduction bands are exhausted for the first valence band. This entire sequence repeats for the next valence band (v_2), and so on, until all valence-conduction combinations are covered.

The total number of rows corresponds to the size of the basis set:

$$\text{Total Rows} = N_v \times N_c \times \text{Total } k \text{ points}$$

Example file:

```
%--   Exciton State S=1       Exciton State S=2    ...   Index
    1.308e-07 -9.886e-08i   1.381e-06 -1.138e-07i  ...  %(v1c1k1)
    2.919e-08 -8.831e-08i   2.418e-06 -2.237e-05i  ...  %(v1c1k2)
    ...
    5.318e-03 -6.866e-07i   7.937e-01 +4.489e-06i  ...  %(v1c2k1)
    ...
    3.450e-07+ 6.726e-08i   4.381e-06 -1.138e-03i  ...  %(vNvcNck1)
     ...
```

3.3.3 Simulation Log (WBSE_log.txt)

This plain text file serves as the runtime record of the simulation, providing a detailed summary of the computational performance. Its primary purpose is to allow users to assess the computational cost of the calculation in terms of time and resources. The log records the timestamp of completion, the precise dimensions of the system (number of k-points and active bands), the number of CPU cores utilized, and a detailed breakdown of the wall-clock time required for both the matrix construction and diagonalization phases. This breakdown is particularly useful for benchmarking performance and optimizing parallelization settings.

Example file:

```
This calculation was finished at 14-Jul-2025 10:15:35

-------------- k-points and bands and included in BSE -------------
number of k-points Nk = 2400
```

```
numbers of valence and conduction bands (Nv,Nc) = ( 2 , 2 )


-------------- CPU usage --------------
The number of cores used in the construction of BSE matrix = 6

-------------- Computation time --------------
Computation time of constructing  BSE matrix  =        70.368 sec.
Computation time of diagonalizing BSE matrix  =        50.058 sec.
                      Total computation time  =       120.426 sec.
```

3.3.4 Momentum Grid Data: k.mat

This MATLAB binary file stores the explicit coordinates of the momentum space grid used in the calculation. It contains the exact same vector data as the text-based kmesh.txt file but is saved in a binary format to facilitate direct and efficient loading during post-processing. The data is stored as a double-precision matrix with dimensions $N_k \times 3$. Each row corresponds to a unique $\boldsymbol{k}$-point index i. The three columns represent the Cartesian componentsk_x, k_y, and k_z (in Å^{-1}). Consistent with the validation and sorting logic described in Section 3.2.1, the grid points in this file are organized so that the dense region is listed first, followed by the coarse region. For detailed information regarding the grid generation logic and the distribution of points, please refer to the Momentum Space Grid (kmesh.txt) description in Section 3.3.1.

Example file:

```
      kx              ky              kz
-0.3289873897   0.6078110657   0.0000000000   % k1
-0.2960886507   0.6647933531   0.0000000000   % k2
-0.3289873897   0.6457992573   0.0000000000   % k3
      ...             ...             ...
```

3.3.5 Integration Weights: wi.mat

This MATLAB binary file stores the integration weights (w_i) associated with the momentum grid. These weights are mathematically derived from the discretization of the Bethe-Salpeter integral equation into a matrix eigenvalue problem. In the numerical implementation, the continuous integration over the Brillouin zone is replaced by a summation over discrete $\boldsymbol{k}$-points:

$$\int \frac{d^2k}{(2\pi)^2} \cdots \rightarrow \sum_i \frac{w_i}{(2\pi)^2} \cdots$$

Geometrically, each $\boldsymbol{k}$-point in the mesh represents the centroid of a specific area element, typically a small triangle within the D_{3h} symmetry mesh. The value w_i corresponds to the area of this Voronoi cell or triangle occupied by the i-th $\boldsymbol{k}$ -point.

These weights are essential for properly normalizing the exciton wavefunctions and for computing optical matrix elements.

Regarding the file structure, the data is stored as a double-precision column vector with dimensions $\text{total number of } \boldsymbol{k} - \text{points} \times 1$. The ordering of the rows strictly corresponds to the sequence of $\boldsymbol{k}$ -points listed in k.mat and kmesh.txt, such that the value in the i-th row represents the specific weight for the coordinate found in the i-th row of the grid files.

Example file:

```
0.001874645399843
0.001874645399843
0.001874645399843
0.001874645399843
0.001874645399843
...
```

3.3.6 Rearranged Band Data: TB_Bands/

This sub-directory is a specialized repository used for research traceability. It is generated only when a user manually provides conduction or valence band files (c*_TB.txt or v*_TB.txt) in the User_input directory.

As discussed in Section 3.2.2, the software automatically reorganizes manual input files to ensure the k-point sequence matches the internal dense-to-coarse sorting required for numerical stability. To ensure the calculation remains transparent, WannierBSE saves the resulting "synchronized" versions of these files here. It is important to note that the internal structure of these files, including the format for eigenvalues and wavefunction coefficients, remains exactly as detailed in Section 3.2.2.

- Content: Contains rearranged versions of the user-provided .txt band files.
- Condition: This folder will not be created if the software generates the bands internally, as those results are stored in the Precomputed_data folder.
- Usage: Users can refer to these files to verify the exact alignment of eigenvalues and eigenvectors used in the final BSE diagonalization.

4. Examples

This chapter illustrates the capabilities of WannierBSE through three distinct tutorials. The examples included in the distribution (located in the examples directory) are chosen to demonstrate the code's workflow-agnostic design and its flexibility in handling different levels of approximation.

WannierBSE allows for "Total Flexibility" in how you approach the Bethe-Salpeter Equation. You may rely on the standard Wannier90 workflow for accurate *ab initio* band structures, or "Bring Your Own Tight-Binding Model" to test analytical models. Similarly, you have the choice of providing a simple, user-defined dielectric constant or utilizing the internal Poisson solver to capture complex environmental screening effects in 2D materials.

The following sections guide you through these different operational modes.

4.1 Example 1: MoS_2 Heterostructure (hBN/MoS_2/SiO_2)

Directory: 01_hBN_MoS2_SiO2

This example combines the *ab initio* accuracy of Wannier90 with the environmental screening simulated by the WannierBSE Poisson solver. We simulate a realistic experimental setup: a monolayer of MoS_2 supported by a Silicon Dioxide (SiO_2) substrate and covered by hexagonal Boron Nitride (hBN).

Key Input Files:

- wannier90_hr.dat: In this case the user is only requested to provide the tight-binding Hamiltonian generated by a standard Wannier90 calculations.

Parameter settings:

- WTB_control.txt: The top valence and the lowest conduction bands corresponding to the wannier90_hr.dat file are $\mathrm{VBM} = 14$ and $\mathrm{CBM} = 15$.
- structure.txt: Contains the lattice vectors and atomic positions of the WSe_2 unit cell, matching the geometry used to generate wannier90_hr.dat.
- kmesh_control.txt: The grid generation uses Nk_coarse=8 to set the base resolution along the Γ to K line. The high-density coverage is defined by Dense_range=2 (in coarse-grid units), while the sampling density within those regions is set by Nk_dense=16, representing the number of k-points along a single direction in the dense sub-mesh.
- control.txt: The BSE Hamiltonian is constructed using 2 valence and 2 conduction bands (Nv=2, Nc=2). The solver is configured to compute the lowest 100 excitonic eigenvalues (NVAL=100).
- WF_centers.txt: The corresponding index I of each atomic center τ_I where the i-th Wannier function is localized, used during the Wannier90 calculations is in the WF_centers.txt file.

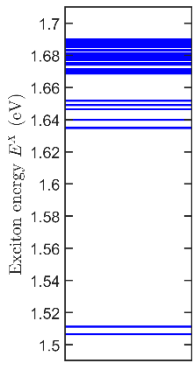
- Dielectric_control.txt: To characterize the dielectric response of the MoS_2 monolayer, the intrinsic material properties are defined by a plasmon peak energy Epl=22.5 eV and in-plane dielectric constant ep_p=13.24. To model the encapsulation of the monolayer between hBN and the SiO_2, the heterostructure is configured with one capping layer and one substrate layer (t_layers=s_layers=1) characterized by dielectric constants of t_dielectric = 5.89 and s_dielectric= 3.9 respectively. The vertical geometry is established via layer_boundaries = 3.13, -3.13 (Å), which positions the hBN/MoS_2 and MoS2/SiO_2 interfaces maintaining consistency with the monolayer thickness d=6.26 Å specified in the control.txt file.

Exciton data:

Once the input files are prepared and the parameters are configured, the calculation is initiated by executing the main routine.
Upon running the WBSE.m main script, the code constructs the BSE Hamiltonian and performs the diagonalization. The resulting excitonic properties are saved to the following output files:

- Ex.mat: This file contains the calculated exciton eigenvalues (in eV), ordered from lowest to highest. To visualize these results, use the auxiliary script Ex_plot.m, which automatically processes the .mat data to generate the energy spectrum plot shown below.

	1	
1	1.5064	E_1^X
2	1.5064	E_2^X
3	1.5112	E_3^X
4	1.5112	E_4^X
5	1.6348	E_5^X
6	...	



- A.mat: Stores the exciton wavefunctions (eigenvectors). Each column in this matrix represents the expansion coefficients of the excitonic state corresponding to the energies listed in Ex.mat.

	$A_1(vc\boldsymbol{k})$ 1	$A_2(vc\boldsymbol{k})$ 2	
1	0.0024 + 0.0023i	-3.032e-06 + 3.112e-06i	$v_1c_1\boldsymbol{k}_1$
2	0.0022 + 0.0023i	-3.013e-06 + 3.081e-06i	$v_1c_1\boldsymbol{k}_2$
3	0.0032 + 0.0032i	-2.910e-06 + 2.745e-06i	$v_1c_1\boldsymbol{k}_3$
4	0.0026 + 0.0029i	-2.979e-06 + 2.888e-06i	$v_1c_1\boldsymbol{k}_4$
5	0.0022 + 0.0030i	-2.943e-06 + 2.808e-06i	$v_1c_1\boldsymbol{k}_5$
6	...	...	

4.2 Example 2: MoS_2 in a complex Heterostructure (air/hBN/MoS_2/ice/SiO_2)

Directory: 02_air_hBN_MoS2_ice_SiO2

In this example, we extend the Poisson solver's application to a multi-layered environment. This setup accounts for the hydrophilic nature of certain substrates, where an interfacial ice layer may form between the TMD and the support.

Specifically, we simulate a monolayer of MoS_2 on a Silicon Dioxide (SiO_2) with an intermediate 10 Å ice layer. The sample is capped with a hBN 10 nm layer and a semi-infinite air top-layer.

Key Input Files:

- wannier90_hr.dat: In this case the user is only requested to provide the tight-binding Hamiltonian generated by a standard Wannier90 calculations.

Parameter settings:

- WTB_control.txt: The top valence band and to lowest conduction bands, corresponding to the wannier90_hr.dat file, are $\mathrm{VBM} = 14$ and $\mathrm{CBM} = 15$.
- structure.txt: Contains the lattice vectors and atomic positions of the WSe_2 unit cell, matching the geometry used to generate wannier90_hr.dat.
- kmesh_control.txt: The grid generation uses Nk_coarse=8 to set the base resolution along the Γ to K line. The high-density coverage is defined by Dense_range=2 (in coarse-grid units), while the sampling density within those regions is set by Nk_dense=16, representing the number of k-points along a single direction in the dense sub-mesh.
- control.txt: The BSE Hamiltonian is constructed using 2 valence and 2 conduction bands (Nv=2, Nc=2). The solver is configured to compute the lowest 100 excitonic eigenvalues (NVAL=100).
- WF_centers.txt: The corresponding index I of each atomic center τ_I where the i-th Wannier function is localized, used during the Wannier90 calculations is in the WF_centers.txt file.
- Dielectric_control.txt: The dielectric response of the MoS_2 monolayer is defined by its intrinsic material properties, using a plasmon peak energy Epl=22.5 eV and in-plane dielectric constant of ep_p=13.24.
  To represent the complex environmental screening of this heterostructure, the system is configured with two capping layers (air and hBN) and two substrate layers (ice and SiO_2), set via t_layers=s_layers=2, respectively. Ordered from the top-most layer to the bottom-most, the dielectric constants are defined as t_dielectric = 1, 5.89 and s_dielectric=1.9, 3.9.
  The vertical geometry is established via layer_boundaries = 103.13, 3.13, -3.13, 13.13 (Å). These coordinates define the interfaces for air/hBN, hBN/MoS_2, MoS2/ice, and ice/SiO_2. This configuration ensures consistency with a 10 nm

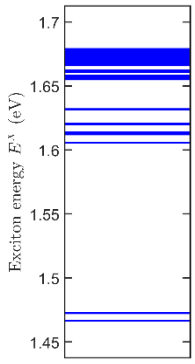
hBN capping layer, a 10 Å interfacial ice layer, and an effective monolayer thickness of d=6.26 Å, which is specified in the control.txt file.

Exciton data:

Once the input files are prepared and the parameters are configured, the calculation is initiated by executing the main routine.
Upon running the WBSE.m main script, the code constructs the BSE Hamiltonian and performs the diagonalization. The resulting excitonic properties are saved to the following output files:

- Ex.mat: This file lists the calculated exciton energies (in eV), ordered from lowest to highest.

	1	
1	1.4666	E_1^X
2	1.466	E_2^X
3	1.4725	E_3^X
4	1.4725	E_4^X
5	1.6057	E_5^X
6	...	



- A.mat: As described in the previous section, this file stores the exciton eigenvectors. The matrix represents the expansion coefficients of the exciton states within the transition basis (v, c, k).

	$A_1(vc\boldsymbol{k})$ 1	$A_2(vc\boldsymbol{k})$ 2	
1	-0.0018 - 0.0023i	9.751e-07 + 4.333e-06i	$v_1c_1\boldsymbol{k}_1$
2	-0.0016 - 0.0023i	9.547e-07 + 4.299e-06i	$v_1c_1\boldsymbol{k}_2$
3	-0.0024 - 0.0033i	7.339e-07 + 4.133e-06i	$v_1c_1\boldsymbol{k}_3$
4	-0.0019 - 0.0030i	8.110e-07 + 4.222e-06i	$v_1c_1\boldsymbol{k}_4$
5	-0.0015 - 0.0030i	7.547e-07 + 4.144e-06i	$v_1c_1\boldsymbol{k}_5$
6	...	...	

4.3 Example 3: MoS_2 using an Analytic Tight-Binding Model

Directory: 03_MoS2_analytic_TB_model

This example demonstrates the code's ability to interface with external or analytical models, bypassing Wannier90 entirely. As example, we calculate the excitonic spectrum of Monolayer MoS2 using the two-band tight-binding model described by Berkelbach *et al.* [20].

Here, we utilize the internal Poisson solver to compute the dielectric screening, assuming the monolayer is suspended in a vacuum (Air/MoS_2/Air). This example highlights how users can provide their own Hamiltonian matrix elements and k-point meshes directly.

Key Input Files:

The user shall provide the primary input files:

- c1_TB.txt and v1_TB.txt: User-supplied energy dispersion files for the conduction and valence bands.
- kmesh.txt: The corresponding coordinate mesh in reciprocal space used in the conduction and valence band files

Parameter settings:

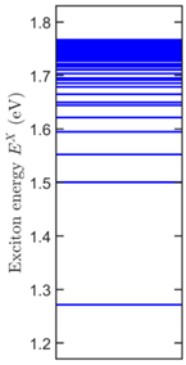
- structure.txt: Contains the lattice vectors and atomic positions of the MoS_2 monolayer unit cell, matching the geometry used to generate the conduction and valence bands.
- control.txt: The MoS_2-ML thickness is set $d = 6.26\ (\text{Å})$. The BSE Hamiltonian is constructed using 2 valence and 2 conduction bands (Nv=1, Nc=1). The solver is configured to compute the lowest 100 excitonic eigenvalues (NVAL=100).
- WF_centers.txt: Following the 2-band model described by Berkelbach et al., the Hamiltonian basis is constructed from two orbitals localized on the transition-metal atom ($I = 1$).These include the $|d_{z^2}\rangle \equiv |\phi_c\rangle$ orbital for the conduction band and the $|d_{x^2-y^2}\rangle + i\tau|d_{xy}\rangle \equiv |\phi_v^\tau\rangle$ orbital for the valence band.
- Dielectric_control.txt: To calculate the dielectric function of a suspended MoS_2 monolayer, the material is characterized by a plasmon peak energy Epl = 22.5 eV and an in-plane dielectric constant ep_p = 13.24. To simulate the vacuum or air environment surrounding the monolayer, the system is configured with a single capping and substrate layer (t_layers = 1, s_layers = 1) both assigned a dielectric constant of t_dielectric = s_dielectric = 1.0. In accordance with a monolayer thickness of d = 6.26 Å, the air/MoS_2 and MoS_2/air interfaces are symmetrically positioned via layer_boundaries = 3.13, -3.13 (Å).

Exciton data:

Similarly to previous example, upon running the main script WBSE.m main, the code constructs the BSE Hamiltonian and performs the diagonalization. The resulting excitonic properties are saved to the following output files:

- Ex.mat: This file lists the calculated exciton energies (in eV), ordered from lowest to highest.

	1	
1	1.2717	E_1^X
2	1.5003	E_2^X
3	1.5006	E_3^X
4	1.5524	E_4^X
5	1.5948	E_5^X
6	...	



- A.mat: As described in the previous section, this file stores the exciton eigenvectors. The matrix represents the expansion coefficients of the exciton states within the transition basis (v, c, k).

	$A_1(vc\boldsymbol{k})$	$A_2(vc\boldsymbol{k})$	
	1	2	
1	-0.0038 - 0.0117i	1.326e-05 + 6.911e-07i	$v_1c_1\boldsymbol{k_1}$
2	-0.0040 - 0.0117i	-1.327e-05 - 5.007e-07i	$v_1c_1\boldsymbol{k_2}$
3	-0.0037 - 0.0120i	5.606e-05 + 4.127e-06i	$v_1c_1\boldsymbol{k_3}$
4	-0.0038 - 0.0118i	2.788e-05 + 1.654e-06i	$v_1c_1\boldsymbol{k_4}$
5	-0.0041 - 0.0117i	-2.792e-05 - 8.524e-07i	$v_1c_1\boldsymbol{k_5}$
6	...	...	

4.4 Example 4: WSe_2 with Constant Dielectric Function

Directory: 04_WSe2_constant_epsilon

This example demonstrates how WannierBSE allows users to integrate their own custom dielectric models. Here, we apply this flexibility to study a freestanding WSe_2 monolayer by adopting a constant effective dielectric screening, implemented within the framework of the local screening approximation.

In this configuration, the internal Poisson solver is disabled. Instead, the user manually defines the screening environment via the epsilon.txt file.

Key Input Files:

The calculation relies on the following primary input files:

- wannier90_hr.dat: The tight-binding Hamiltonian generated by a standard Wannier90 calculation.
- epsilon.txt: A user-provided file defining the dielectric function base on his own model used to screen the Coulomb interaction, in this case a constant effective dielectric constant $\epsilon_{eff} = 4.39$.

Parameter settings:

A summary of the parameter's configuration set in for this example is provided below:

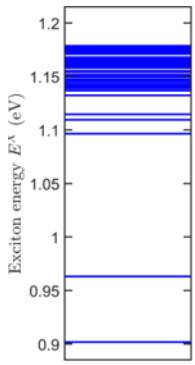
- WTB_control.txt: The top valence band and to lowest conduction band corresponding to the wannier90_hr.dat are $\mathrm{VBM} = 14$ and $\mathrm{CBM} = 15$.
- structure.txt: Contains the lattice vectors and atomic positions of the WSe_2 unit cell, matching the geometry used to generate wannier90_hr.dat.
- kmesh_control.txt: The grid generation uses Nk_coarse=8 to set the base resolution along the Γ to K line. The high-density coverage is defined by Dense_range=2 (in coarse-grid units), while the sampling density within those regions is set by Nk_dense=16, representing the number of k-points along a single direction in the dense sub-mesh.
- WF_centers.txt: The corresponding index I of each atomic center τ_I where the i-th Wannier function is localized, used during the Wannier90 calculations is in the WF_centers.txt file.
- control.txt: The BSE Hamiltonian is constructed using 2 valence and 2 conduction bands (Nv=2, Nc=2). The solver is configured to compute the lowest 100 excitonic eigenvalues (NVAL=100).

Exciton data:

Following the same procedure as the previous example, executing the main script WBSE.m initiates the construction and diagonalization of the BSE Hamiltonian. The calculated excitonic properties are subsequently stored in the following output files:

- Ex.mat: This file lists the calculated exciton energies (in eV), ordered from lowest to highest.

	1	
1	0.9017	E_1^X
2	0.9017	E_2^X
3	0.9631	E_3^X
4	0.9631	E_4^X
5	1.0965	E_5^X
6	...	



- A.mat: As described in the previous section, this file stores the exciton eigenvectors. The matrix represents the expansion coefficients of the exciton states within the transition basis (v, c, k).

	$A_1(vc\boldsymbol{k})$	$A_2(vc\boldsymbol{k})$	
	1	2	
1	-14.0212 + 48.3443i	0.0111 + 0.0069i	$v_1c_1\boldsymbol{k_1}$
2	-14.1458 + 48.3080i	0.0111 + 0.0069i	$v_1c_1\boldsymbol{k_2}$
3	-12.3469 + 43.0594i	0.0099 + 0.0061i	$v_1c_1\boldsymbol{k_3}$
4	-13.1492 + 45.4817i	0.0104 + 0.0064i	$v_1c_1\boldsymbol{k_4}$
5	-13.3434 + 45.4251i	0.0104 + 0.0065i	$v_1c_1\boldsymbol{k_5}$
6	...	...	

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